

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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Date: 02-08-2026

1. TITLE

Develop a College Information Portal Using HTML5

2. AIM

To develop a static web page — the **College Information Portal** — using **HTML5**, demonstrating semantic elements, multimedia embedding, HTML5 form input types, lists, and tabular data.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basics of HTML5 and its semantic elements.
 - To create a structured web page using `<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, and `<footer>`.
 - To embed multimedia using `<img>`, `<audio>`, and `<video>` tags.
 - To build a registration form using HTML5 input types: text, email, password, date, number, color, range, radio, checkbox, select, and textarea.
 - To display tabular data using `<table>`, `<thead>`, `<tbody>`, `<tr>`, `<th>`, and `<td>`.
 - To apply inline text formatting using `<b>`, `<i>`, `<u>`, `<sub>`, and `<sup>`.
-

4. THEORY

HTML5 is a JavaScript runtime environment that allows JavaScript code to execute outside the browser.

Semantic Elements such as `<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, and `<footer>` give meaning to the structure of the page, improving readability and accessibility.

Multimedia Tags — `<audio>` and `<video>` — allow embedding of audio and video content directly in the browser.

In this page, HTML5 forms are used to collect student registration data.

- type="email" validates the email format in the browser.
- type="date" provides a built-in date picker.
- type="number" restricts input to numeric values within a range.
- type="color" opens a colour picker widget.
- type="range" renders a slider control.
- type="radio" allows selection of exactly one option from a group.
- type="checkbox" allows multiple selections from a group.

The basic structure of the page is:

Header → Navigation → Main Content → Footer

5. ALGORITHM / PROCEDURE

- Create a new file named College_Information_Portal.html.
- Add the <!DOCTYPE html> declaration.
- Add the <html lang="en"> root element.
- Inside <head>, add charset meta tag, viewport meta tag, and the page title.
- Create a <header> with a layout table containing the college logo, portal heading, and navigation link.
- Add a <hr> separator after the header.
- Inside <main>, create the "About Our Institution" <section> with an <article>.
- Use , <i>, <u>, <sub>, and <sup> for inline formatting within the article.
- Add an unordered list for Key Academic Departments.
- Add an ordered list for the Admission Procedure steps.
- Create the "Campus Gallery & Media" <section> with , <audio>, and <video> tags.
- Create the "Student Registration Form" <section> with a <form action="#" method="post">.
- Inside the form add HTML5 input types: text, email, password, date, number, color, range, radio, checkbox, select, textarea.
- Create the "Student Details" <section> with a bordered <table> and three student rows.
- Add a <footer> with a copyright notice.
- Open the file in a browser and verify the output.

6. PROGRAM

College_Information_Portal.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
```

```
</head>
<body>
  <!-- Semantic Header Element -->
  <header>
    <table width="100%">
      <tr>
        <td width="40%">
          
        </td>
        <td width="30%">
          <!-- College Logo with Alternate Text -->
          <h1>College Information Portal</h1>
        </td>
        <td width="40%">
          <!-- Semantic Navigation Element -->
          <nav>
            <p>
              <a href="https://www.anits.org/">Official College Website</a>
            </p>
          </nav>
        </td>
      </tr>
    </table>
  </header>
  <hr>
  <!-- Main Content Area -->
  <main>
    <!-- Basic HTML & Semantic Article Section -->
    <section>
      <center>
        <h2>About Our Institution</h2>
      </center>
      <article>
        <h3>Overview</h3>
        <p>
```

Welcome to the College Information Portal.

Our institution is dedicated to <i>academic excellence</i>.

We offer <u>modern campus facilities</u>.

</p>

<p>

Note: Chemistry experiments require H₂O solutions,
and our innovation cell secured 1st place.

</p>

</article>

<!-- Unordered List -->

<h3>Key Academic Departments</h3>

Computer Science & Engineering

Electronics & Communication Engineering

Mechanical Engineering

Civil Engineering

<!-- Ordered List -->

<h3>Admission Procedure</h3>

Complete the online application form.

Submit required academic transcripts.

Attend the counselor interview.

Confirm fee payment and final enrollment.

</section>

<hr>

<!-- Media Section (Images, Audio & Video) -->

<table width="100%">

<tr>

<td width="50%" valign="top">

<section>

<h2>Campus Gallery & Media</h2>

<h3>Campus View</h3>

<h3>Campus Audio Announcement</h3>

<audio controls>

```

        <source src="announcement.mp3" type="audio/mpeg">
        Your browser does not support the audio element.
    </audio>
    <h3>Campus Tour Video</h3>
    <video controls width="320">
        <source src="tour.mp4" type="video/mp4">
        Your browser does not support the video element.
    </video>
</section>
</td>
<td>
    <section>
        <!-- Forms & HTML5 Input Types -->
        <h2>Student Registration Form</h2>
        <form action="#" method="post">
            <fieldset>
                <legend> Register </legend>
                <label for="fullname">Full Name:</label>
                <input type="text" id="fullname" name="fullname" required>
                <!-- HTML5 Input Type: Email -->
                <label for="email">Email Address:</label>
                <input type="email" id="email" name="email" required>
                <label for="password">Password:</label>
                <input type="password" id="password" name="password" required>
                <!-- HTML5 Input Type: Date -->
                <label for="dob">Date of Birth:</label>
                <input type="date" id="dob" name="dob">
                <!-- HTML5 Input Type: Number -->
                <label for="age">Age:</label>
                <input type="number" id="age" name="age" min="16" max="100">
                <!-- HTML5 Input Type: Color -->
                <label for="fav_color">Preferred Portal Theme Color:</label>
                <input type="color" id="fav_color" name="fav_color">
                <!-- HTML5 Input Type: Range -->
                <label for="skill_level">Self-Assessed Tech Skill Level (0-100):</label>
                <input type="range" id="skill_level" name="skill_level" min="0"
max="100">

```

```

        <label>Gender:</label>
        <input type="radio" id="male" name="gender" value="Male">
        <label for="male">Male</label>
        <input type="radio" id="female" name="gender" value="Female">
        <label for="female">Female</label>
        <input type="radio" id="other" name="gender" value="Other">
        <label for="other">Other</label>
        <label>Hobbies:</label>
        <input type="checkbox" id="reading" name="hobbies"
value="Reading">
        <label for="reading">Reading</label>
        <input type="checkbox" id="sports" name="hobbies" value="Sports">
        <label for="sports">Sports</label>
        <input type="checkbox" id="coding" name="hobbies" value="Coding">
        <label for="coding">Coding</label>
        <label for="branch">Branch Selection:</label>
        <select id="branch" name="branch">
            <option value="CSE">Computer Science</option>
            <option value="ECE">Electronics & Communication</option>
            <option value="EEE">Electrical & Electronics</option>
            <option value="ME">Mechanical Engineering</option>
            <option value="CE">Civil Engineering</option>
        </select>
        <label for="comments">Comments:</label>
        <textarea id="comments" name="comments" rows="4"
cols="40"></textarea>
        <input type="submit" value="Submit">
        <input type="reset" value="Reset">
    </fieldset>
</form>
</section>
</td>
</tr>
</table>
<hr>
<!-- Tables -->
<section>

```

```
<center>
  <h2>Student Details</h2>
  <table border="1">
    <thead>
      <tr>
        <th>Roll No</th>
        <th>Name</th>
        <th>Branch</th>
        <th>Email</th>
      </tr>
    </thead>
    <tbody>
      <tr>
        <td>101</td>
        <td>Alice Johnson</td>
        <td>Computer Science</td>
        <td>alice@example.com</td>
      </tr>
      <tr>
        <td>102</td>
        <td>Bob Smith</td>
        <td>Mechanical Engineering</td>
        <td>bob@example.com</td>
      </tr>
      <tr>
        <td>103</td>
        <td>Charlie Davis</td>
        <td>Electrical Engineering</td>
        <td>charlie@example.com</td>
      </tr>
    </tbody>
  </table>
</center>
</section>
<br><br>
<hr>
</main>
```

```
<hr>
<!-- Semantic Footer Element -->
<footer>
  <center>
    <p>&copy; 2026 College Information Portal. All rights reserved.</p>
  </center>
</footer>
</body>
</html>
```

7. CODE EXPLANATION

Code	Explanation
<html>	Defines the HTML document.
<head>	Contains webpage information and title.
<body>	Contains the visible webpage content.
<header>	Defines the header of the portal.
	Displays the college logo or campus image.
<h1>	Defines the main heading.
<nav>	Defines the navigation section.
<a>	Creates a link to the college website.
<main>	Contains the main content of the portal.
<section>	Groups related content together.
<article>	Contains the college overview information.
/ 	Creates unordered and ordered lists.
<audio>	Adds audio to the webpage.
<video>	Adds video to the webpage.
<form>	Creates the student registration form.
<input>	Accepts different types of student information.
<select>	Creates a dropdown for branch selection.
<textarea> >	Allows users to enter comments.
<table>	Displays student details in tabular form.
<th>	Defines table headings.
<td>	Defines table data cells.
<footer>	Defines the footer of the portal.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Open the HTML file in a web browser	Static webpage should be displayed correctly	Static webpage displayed correctly	Pass
TC-02	Click the Home navigation link	Home section should be displayed	Home section displayed	Pass
TC-03	Click Articles and About links	Corresponding sections should be displayed	Corresponding sections displayed	Pass
TC-04	Enter valid name, email, topic and message in the form	Form should accept the entered information	Information accepted	Pass
TC-05	Click Reset Entries	All form fields should be Reset	All form fields are Reset	Pass
TC-06	Leave required fields empty and submit	Browser should display required-field validation	Validation displayed	Pass

9. OUTPUT

Output 1: Before Student Registration



About Our Institution

Overview

Welcome to the **College Information Portal**. Our institution is dedicated to *academic excellence* and world-class research. We offer modern campus facilities to nurture overall student development.

Note: Chemistry experiments require H₂O solutions, and our innovation cell secured 1st place in national rankings.

Key Academic Departments

- Computer Science & Engineering
- Electronics & Communication Engineering
- Mechanical Engineering
- Civil Engineering

Admission Procedure

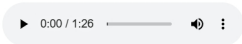
- Complete the online application form.
- Submit required academic transcripts.
- Attend the counselor interview.
- Confirm fee payment and final enrollment.

Campus Gallery & Media

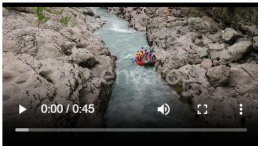
Campus View



Campus Audio Announcement



Campus Tour Video



Student Registration Form

Register

Full Name:

Email Address:

Password:

Date of Birth:

dd-mm-yyyy

Age:

Preferred Portal Theme Color:

Self-Assessed Tech Skill Level (0-100):

Gender:

☐ Male

☐ Female

☐ Other

Hobbies:

☐ Reading

☐ Sports

☐ Coding

Branch Selection:

Computer Science

Comments:

Submit

Reset

Student Details

Roll No	Name	Branch	Email
101	Alice Johnson	Computer Science	alice@example.com
102	Bob Smith	Mechanical Engineering	bob@example.com
103	Charlie Davis	Electrical Engineering	charlie@example.com

Output 2: After Student Registration



About Our Institution

Overview

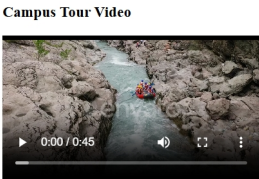
Welcome to the **College Information Portal**. Our institution is dedicated to *academic excellence* and world-class research. We offer modern campus facilities to nurture overall student development.

Note: Chemistry experiments require H₂O solutions, and our innovation cell secured 1st place in national rankings.

- Key Academic Departments**
- Computer Science & Engineering
 - Electronics & Communication Engineering
 - Mechanical Engineering
 - Civil Engineering

- Admission Procedure**
1. Complete the online application form.
 2. Submit required academic transcripts.
 3. Attend the counselor interview.
 4. Confirm fee payment and final enrollment.

Campus Gallery & Media



Student Registration Form

Register

Full Name:

Purna

Email Address:

Example@gmail.com

Password:

Date of Birth:

10-05-2006

Age:

20

Preferred Portal Theme Color:

Self-Assessed Tech Skill Level (0-100):

Gender:

☒ Male ☐ Female ☐ Other

Hobbies:

☒ Reading ☐ Sports ☐ Coding

Branch Selection:

Computer Science

Comments:

Hello

Submit

Reset

Student Details

Roll No	Name	Branch	Email
101	Alice Johnson	Computer Science	alice@example.com
102	Bob Smith	Mechanical Engineering	bob@example.com
103	Charlie Davis	Electrical Engineering	charlie@example.com

10. RESULT

Thus, a structured static webpage was successfully designed and developed using HTML5 semantic elements and HTML components. The webpage was verified for its structure, navigation links, form controls, lists, sidebar and footer.